Unit Two

Disorders of the Respiratory system and Nursing Responsibilities

For third year nursing students

By

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Respiratory System



Review of anatomy & physiology of the respiratory system

- The respiratory system is divided into two major sections:

1. Upper respiratory tract: This consists of :

- nose& Para nasal sinuses
- pharynx
- larynx

* Upper respiratory tract serves as a conducting passage for air as it enters & leaves the lungs.

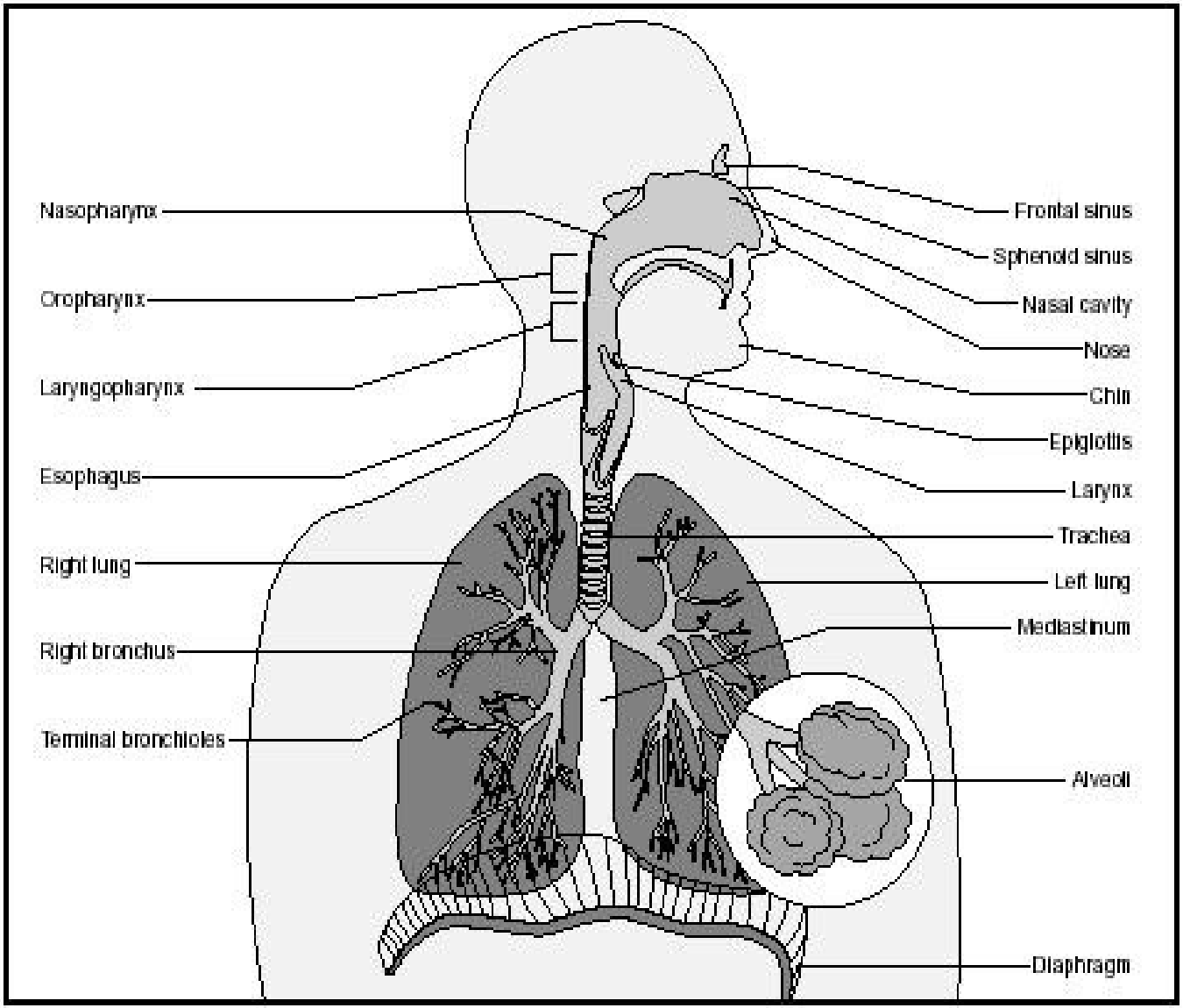
- Its primary functions are to warm, humidify, & filter entering air & to protect the lower airway from foreign material.

2. Lower respiratory tract

* Begins at the trachea or windpipe and ends at the alveoli (air sacs in the lungs where gas exchange takes place). It includes:

- Right & left main stem bronchi / large airways of lungs/
- Segmental & sub segmental bronchi
- Bronchioles
- Alveolar sacs

ANATOMY OF THE RESPIRATORY SYSTEM



1. Nose – air enters body

2. Pharynx – 3 divisions: Nasopharynx, oropharynx, and laryngopharynx

3. Larynx – voice box

4. Glottis – opening to larynx

5. Epiglottis – flap of cartilage that prevents food/fluid from entering lungs

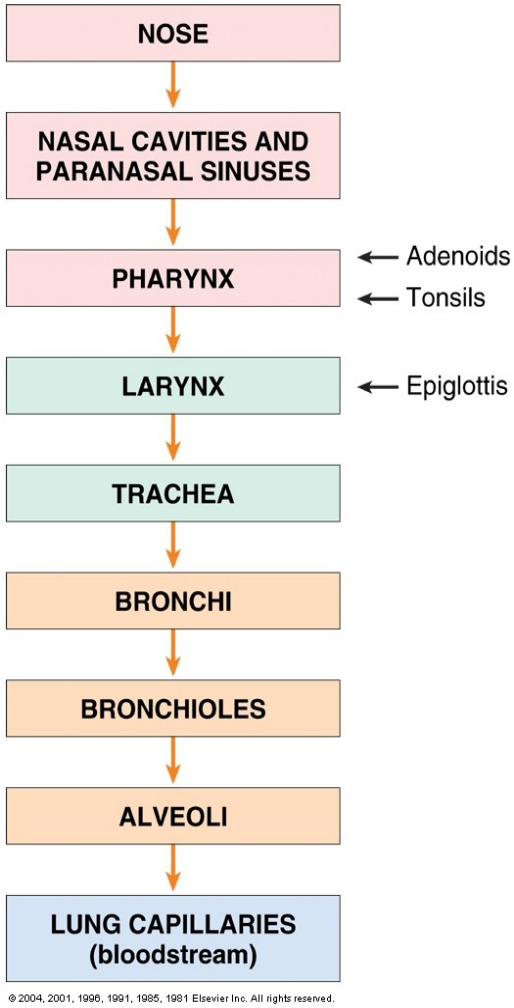
6. Trachea – air passes through on way to lungs

7. Bronchus (bronchi) – tube leading to each lung

8. Bronchioles – smallest bronchial branch

9. Alveolar sacs – air sac at end of bronchioles

10. Gas exchange – occurs at alveolar/capillary level



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Figure Pathway of air from the nose to the capillaries of the lungs.

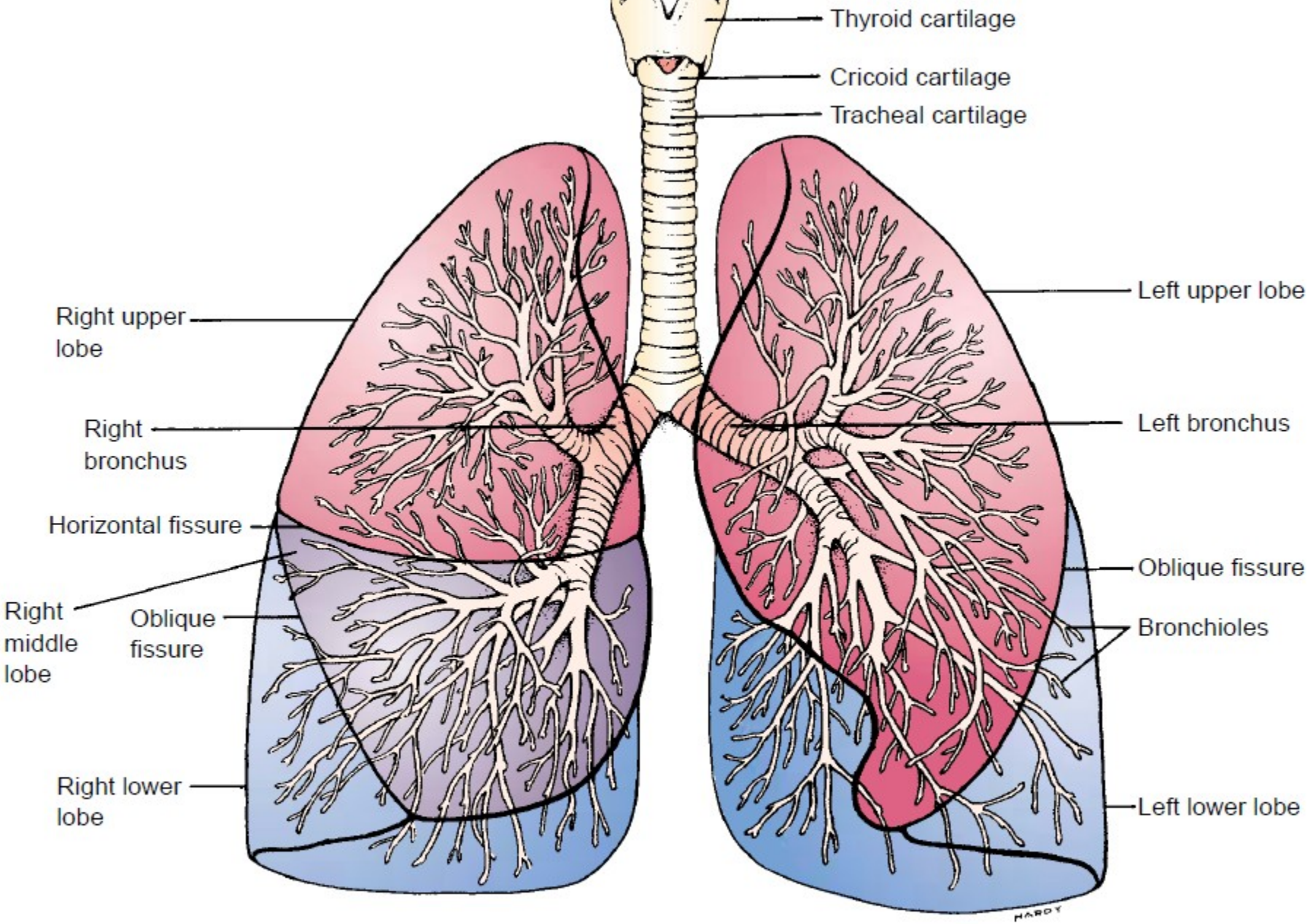


FIGURE 21-4 The lungs consist of five lobes. The right lung has three lobes (upper, middle, lower); the left has two (upper and lower). The lobes are further subdivided by fissures. The bronchial tree, another lung structure, inflates with air to fill the lobes.

* The lungs are covered by two thin membranes:
  A. Visceral pleura: connected to the lungs.
  B. Parietal >> : lining the ribs & chest wall.
* Pleura cavity: is the potential space between visceral & parietal pleura. It contains serous fluid (lubricates, so that it prevent friction rub of the two pleura membrane.
* The lungs are located within the thoracic cavity and it is surrounded by the ribs and muscles.

NURSING ASSESSMENT OF PATIENT WITH RESPIRATORY DISORDER

TAKING A RESPIRATORY HISTORY

- Begin the history by determining the person's_chief complaint.
- Alleviating and aggravating
- Complete information should be obtained to verify the exact nature of major manifestation of broncho-pulmonary disease such as

- **Dyspnea**
- **cough , sputum production,**
- **haemoptysis, &**
- **pain with breathing / pleurtic pain /**

Important terms

- Normal breathing: - A respiratory rate of 14-24 breaths per minute.
- Tachypnea: - Increased respiratory rate over 24 breaths per minute.
- Bradypnea: -Decreased respiratory rate under 14 breaths per minute.
- Apnea:- Total cessation of airflow to the lungs

- Hyperpnea:-Increase in depth of respiration.
- Hyperventilation:-Increase in depth & rate of respiration.
- Hypoventilation: slow or irregular respiratory pattern with shallow respirations.
- Hypoxia =this is situation in which there is lack of oxygen in the body.
- Hypoxaemia:-Oxygen deficiency in the blood.
- Hypercapnia:-is the retention of co_2 in the blood i.e. the Pco2 >45mmHg at rest.

- Chyne-stokes respiration:- A cyclic pattern of progressively deeper respirations , followed by progressively shallow respirations, & a period of apnea
    - e.g. CHF, drug over dose , renal failure

Common diagnostic procedures for respiratory conditions

- **Chest x-ray**

- It is widely used to identify pathologic changes in the lung & chest

- **Bronchoscopy**

- It is direct Visualization & observation of the larynx, trachea, & bronchi through bronchoscope that is designed for passage through trachea.

- **Thoracentesis-** is the insertion of a needle in to a chest under local anesthesia to aspirate excess fluid from pleural cavity.

Purpose:

- Removal of fluid & air from pleural cavity
- Diagnostic aspiration of pleural fluid
- Pleural biopsy
- Instillation of medication in to the pleural cavity.

Lab. tests

1. CBC- to get base line information
2. **Sputum Examination**

- Sputum is obtained to identify pathogenic organism

3. **Arterial blood gas evaluation: Includes:-**

-CO2

-O2

-PH

Nursing interventions for diagnostic procedures

- obtain informed consent
- with hold food & fluids for 6 hours before the test to reduce the risk of aspiration
- Explain the procedure to the patient to reduce fear
- Bring equipments to the room
- Administer prescribed preprocedure medications (usually atropine)
- Relieve anxiety
- Remove contact lenses, dentures, & so on

- After the procedure, the patient is given nothing by mouth until the cough reflex returns
- Monitor respiratory status
- Observe the patient for cyanosis, hypotension, tachycardia, haemoptysis, dyspnea, bronchospasm & blocked air way.
- After the procedure, a pressure dressing may be applied in the case of thoracentesis
- Monitor vital signs frequently.

Upper Respiratory disorders

Tonsillitis

- It is an inflammation & enlargement of the tonsil tissue.

Causative agent:- Group A streptococcus

-It can be viral in origin

-It is very common up to the age of 15years.

Predisposing factors:-URTI

-Lowered immunity

-pollution

Clinical Features:- sore throat, dysphagia, headaches, fever, inflamed tonsil

Diagnostic evaluation

- History
- Physical examination / visual inspection/
- Culture & sensitivity.

Medical Mx:

- Amoxicillin 500mg po. t id for 7 days OR
- Erythromycin 500mg. po. Qid for seven days OR

-Benz. Penicillin 2.4Million lu. Im. stat

Nursing intervention

- Analgesics such as paracetamol 500mg 2 tabs PRN
- Bed rest and soft diet
- Warm saline gargles

Complication: -

- Laryngeal edema
- Acute otitis media
- Septicemia
- Chronic tonsillitis

Chronic tonsillitis

- It is due to recurrent acute tonsillitis attack.

Etiology: - similar to acute tonsillitis

C/M; - recurrent pain

- cough

- enlarged cervical lymph node

R/X: - Nutritious diet & vitamins

- Antibiotic & Analgesics

- Tonsillectomy / for recurrent tonsillitis which does not respond to antibiotic

Pharyngitis

- It is a febrile inflammation of the pharynx.

Causative agent:-

- Viral organism 70%
- Group A streptococcus is the most common bacterial organism.

Clinical Features

– Red pharyngeal membrane & tonsil

– Enlarged & tender cervical lymph nodes

-Fever & sore throat

-adenoiditis-is inflammation of adenoids caused by infection. Adenoids are masses of lymphatic tissue.

Diagnosis: - clinical sign & symptoms

Differential diagnosis: **-**

- tonsillitis
- laryngitis

Complication: -

- Otitis Media
- Mastoditis
- Meningitis

Medical Rx: For bacterial causes

- Ampicillin 500mg po Qid for10 days **or**
- Erythromycin 500mg p.o. QID for 10days
- For viral cause supportive Rx

Nursing Intervention

- Instruct bed rest during febrile stage of illness.
- Liquid or soft diet is provided during acute stage of the disease.
- Warm saline gargles or irrigation are used. Irrigating the throat properly is an effective means of reducing spasm in the pharyngeal muscle.
- Acetaminophen 500mg 2 tabs at 6 hour interval.
- Mouth care can be given

Laryngitis

- It is an inflammation of the larynx often occurs as a result of voice abuse, exposure to dust, chemicals, smoke, & other pollutant.

Etiology: - Almost always virus

- Bacterial invasion may be secondary

Risk factors:

- URTI
- Vocal misuse and over use
- Irritation / smoking, alcohol /
- Seasonal changes
- Iatrogenic / intubations and endo-laryngeal surgery /

- Laryngitis is usually associated with acute rhinitis or naso-pharyngitis.

C/F:- hoarseness or complete loss of voice & severe cough

- pain occurs in severe cases
- stridor may be present in children
- edema, exudates, congestion

- Laryngitis can be the complication of chronic sinusitis & chronic bronchitis.

Medical Management

- Resting the voice
- Avoid smoking
- Bed rest
- Inhaling cool steam
- Treat secondary bacterial infection with antibiotics.

Nursing Intervention

1.Clearing airway:

2. Promote comfort Measures

- Give analgesics / acetaminophen . . . /

3 .Promoting communication

- Voice rest

4. Encouraging fluid intake

5. Patient teaching – treatment regimen, prevention

6. Monitor & Manage potential complication

Lower Respiratory Disorders

Acute tracheo-bronchitis

- Acute tracheobronchitis is an acute inflammation of the mebranes of the trachea & the bronchial tree that often follows infections of the upper respiratory tract.

Causative agents: - it is often virus but the common bacterial causes are:-

- Streptococcus pneumaniae
- Haemephilus Influenza
- Mycoplasma pneumonia

Predisposing factors:-

- Pre- existing URTI
- Inhalation of physical & chemical irritants, gases & other air contaminants

Clinical Manifestations:-

- Initially the patient has a dry, irritating cough & expectorates scanty sputum
- Fever, headache, & generalized malaise.
- As the infection progresses, inspiration may become noisy (inspiratory Stridor) & more profuse purulent sputum may be present.

- Medical Management
  - Advice bed rest.
  - Steam inhalation
  - Advice fluid intake to thin the viscous & tenacious secretions.
  - Dextramethorphan hydro-bromide 15-30mg P.O 3 to 4 times a day.
  - Codeine phosphate 10 – 20 mg P.O. 3-4 times a day

* Antibiotic Treatment:-
  - It is indicated, when bronchitis is complicated by bacterial infections.

- **First line Antibiotics**
- * Ampicillin 500mg P.O Qid, for 7 days. OR
- * Amoxicillin 500gm P.O tid, for 7 days
- **Alternative Antibiotic:-**
- Erythromycin 500mg P.O Qid for 7 days. OR
- Tetracycline 500mg Qid, for 5-7 days. OR
- Co - trimoxazole 480 mg 2 tabs po bid for 7 days

Nursing Interventions:-

- Encourage frequent coughing to remove secretions

– Advice the patient bed rest, steam inhalation & increase fluid intake.

Chronic bronchitis

- It is the presence of a productive cough that lasts 3 months a year for two consecutive years, in the absence of major lung disease (WHO).
- The accumulated secretions in the bronchioles interfere with effective breathing.

Causes: - The major causes are:-

  - Cigarette smoking
  - Exposure to pollution

- Patients with chronic bronchitis are more susceptible to recurring infections of the lower respiratory tract.
- Chronic bronchitis occurs mostly during the winter.

Clinical Manifestations:-

- A chronic, productive cough in the winter months (recurrent coughing and sputum production)
- frequent respiratory infection
- Production of thick, gelatinous sputum (greater amount's produced during super imposed infection)
- Wheezing and dyspnea as disease progresses
- Recurrent acute respiratory infections followed by persistent cough

Diagnostic Evaluation

- A complete history, including family, environmental exposure to irritating substance, & occupational history, and also history of smoking (number of packs per day)
- Physical examination
- Chest x-ray
- Arterial blood gas analysis

Management

- Remove bronchial secretions
- Bronchodilators are prescribed to relieve bronchospasm & reduce air way Obstruction.
- Postural drainage
- Increase fluid intake
- The patient must stop smoking
- Monitor cough, breath sounds, hoarseness, cyanosis, respiratory rate

- Avoid narcotics (cause resp. depression)
- Fowler position to reduce breathing difficulty
- Maintain a calm environment
- Tell the family to humidify the room with boiling water (quickly ease acute croup)
- Warn the family to report
  - Ear ache
  - High fever
  - Productive cough
  - Increased dyspnea immediately

Bronchiectasis

- Bronchiectasis is a chronic abnormal & permanent dilatation of the bronchi and bronchioles

Pathology:-

- The dilatation is due to destructive &inflammatory changes in the wall of medium-sized air ways.
- The normal components of the wall, including cartilage, muscle & elastic tissue, are destroyed &may be replaced by fibrous tissue.

- The dilated airways are frequently contains thick, purulent materials, while peripheral airways are often occluded by secretion or obliterated & replaced by fibrous tissue.

Causes:-

- pulmonary infections
- Obstruction of the bronchus
- Aspiration of foreign bodies & vomitus
- Pressure from tumors, dilated blood vessels & enlarged lymph nodes.

Predisposing factors:-

- Infection in early child hood (measles, influenza, tuberculosis, & immuno-deficiency disorders
- Surgery

- Infection
- Damage bronchia wall

⇓

- Thick sputum production

⇓

- Sputum Obstruct bronchi

⇓

Wall of bronchi will be distended by cough

⇓

- Distended air way.

* *S.aureus & Klebsiella & anaerobes remain important causes of bronchiectasis when antibiotic treatment of pneumonia is not given.*

C/M: - Chronic cough (persistent or recurrent)

- Copious amount of purulent sputum
- Haemoptysis (50-70%)
- P/E: crackles & wheezes may be heard

Diagnosis: - Bronchoscope & CT Scan

- Medical management:-
- Antibiotic based on the results of culture & sensitivity
- Postural drainage
- Avoid smoking & air pollutants
- Plenty of fluid to aid expectoration
- the patient prevent upper respiratory tract infections (take medications properly)

PNEUMONIA

- Pneumonia is an inflammatory process of the lung parenchyma that is commonly caused by infectious agent.

Classification of pneumonia

- There are three classification methods:-

1. Clinical
2. Etiological
3. Radiological

1. **Etiologic classification**

Infectious - are the most common & most important

1/ Bacteria: - can be -

a) Aerobic bacteria

Gram positive	Gram negative
-Streptoccocus	- E. coli
- pneumonia	- Klebsiella
- Staphylococcus aureus	-pseudomonas

b) Anaerobic bacteria

- Bacteriods

2/ Virus

3/ chlamydias

4/ mycoplasms

5/ Riketssiae

2. Radiologic /Anatomic/ Classification

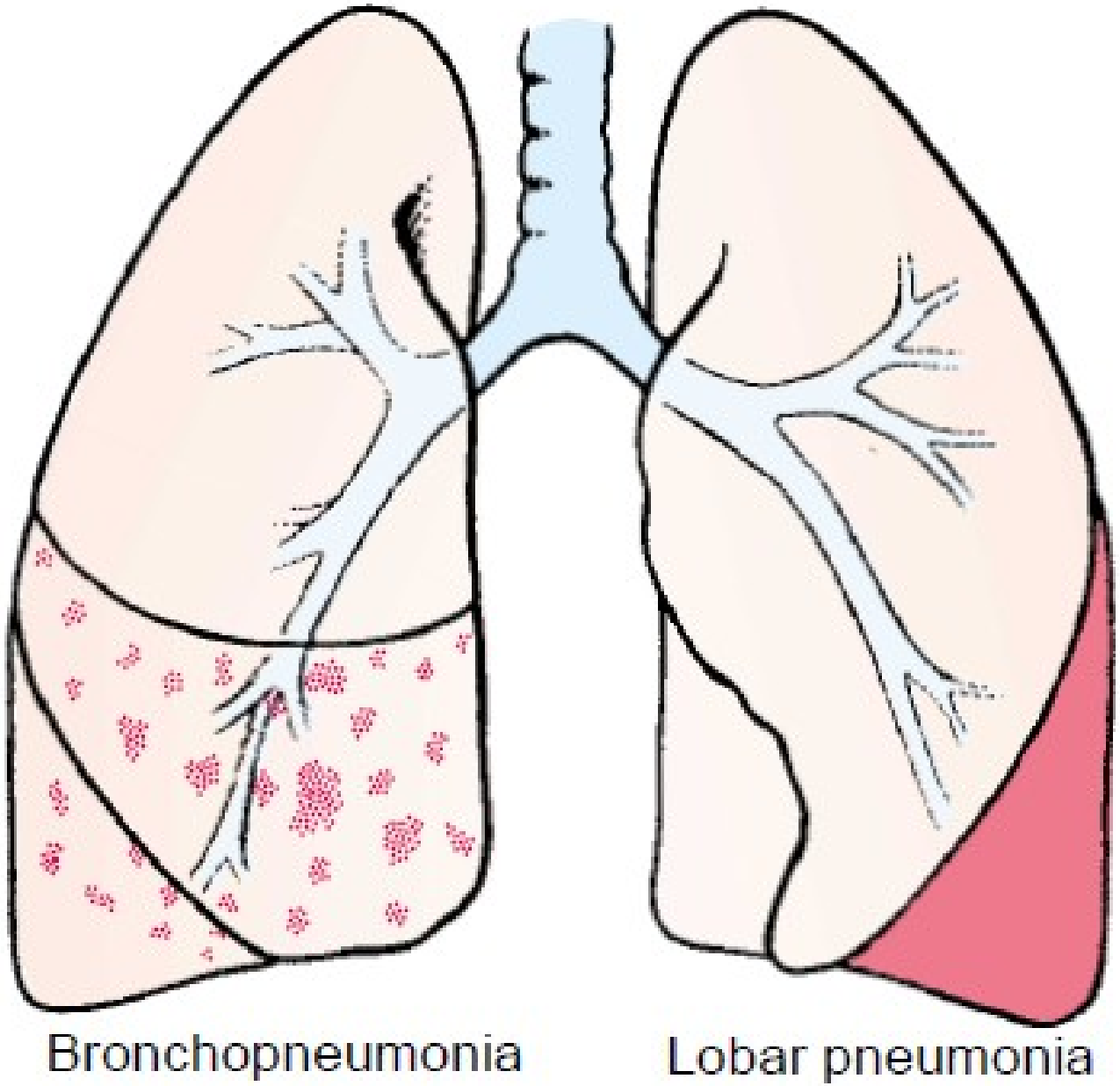
1. Lobar pneumonia :-

- If a substantial portion of one or more lobes is involved or inflamed

2. Broncho Pneumonia:

  - It is distributed in a fashion, having originated in one or more localized areas with in the bronchi & extending to the adjacent surrounding lung parenchyma

- It is more common than lobar pneumonia


Bronchopneumonia
Lobar pneumonia

3. Clinical classification

1. Community Acquired pneumonia / CAP/

2. Hospital Acquired pneumonia / Nosocomial/

3. Aspiration Pneumonia

4. pneumonia in immuno compromised

1. Community Acquired pneumonia / CAP/

- it is a pneumonia in a community.

Causative agent are: -

- Typical or bacteria pneumonia

– streptococcus pneumonia / 50 – 60% /
– staphylococcus aureus
– H. influenzae

- Atypical pneumonia / 20%/

– Ricktesia
– Chlamydia
– Mycoplasmas
– Fungus

2. Hospital Acquired pneumonia /NOSOCOMIAL/ pneumonia

If admitted patient developed pneumonia after 48 hour of admission, it is most likely Nosocomial.

Causative agents: -

- Staph. auerus
- Pseudomonas spp.
- Gram negative enteric bacilli / E. coli/

3. Aspiration pneumonia

- **Predisposing factors for aspiration pneumonia are**: -
  - coma / unconsciousness
  - Seizure.
- **Causative agents**: - most of them are normal flora of oropharynx.
  - They are anaerobic oral spp.
  - They can cause necrosis of lung as a result they cause lung abscess.

Clinical Features: - Mimic pul. Tbc.

- —Fever, chillness
- —Weight loss
- —Pluratic chest pain
- —Blood streaked sputum

4. Pneumonia In immuno compromised patient

Causative agents: -

- Opportunistic / 50%/
  - Pneumocystic carnii, pneumonia / pcp /
  - Cytomegali virus / CMV /
- Bacterial : - those mentioned

➢ Immunity may be suppressed by corticosteroids, chemotherapy, nutritional depletion, AIDS, genetic immune disorders…..

- **Clinical Feature:-**
  - Cough : - first dry then rusty sputum / purulent/ after 24 hours
  - Fever-
  - Pleurtic chest pain / aggravated by inspiration /
  - Tachycardia
  - Bronchial breath sound, dullness & sometimes crepitation

DX: - WBC & differential count

- Sputum gram stain
- Chest x – ray

➢ ***markers of severe pneumonia***

- Respiration rate greater than 30/ min in adult
- Cyanosis
- Hypotension
- Confusion
- If more than or equal to two lobe is involved
- Co – morbidity / asthma, DM. /

Complications

- Para-pneumonic effusion
- Empyema
- Lung abscess
- Lobar collapse

-Sepsis

-Meningitis

-Endocarditis

-Pericarditis

-Arthritis

General Management: -

- Analgesics for pain
- O_2 administration

Specific Management: -

1. Community acquired pneumonia / CAP/

RX: - Amoxicillin 500mg po. Tid 7 days OR

- Ampicillin 500mg po. Qid for 7 days

-Erythromycin 500mg p.o Qid for 7 days

- For severe CAP- Admit the patient
  - Crystalline penicillin 150,000 – 250.000 lu/ kg/ 24 hours, divided in to 6 doses.

** Erythromycin covers all causes of CAP.

2. Hospital acquired pneumonia

*For gram negative bacteria: - Gentamycin 80mg iv / im tid for 7 – 10 days

- For gram positive bacteria (s.aureus): - cloxacillin 500mg po / iv qid for 7 – 10 days

3. Aspiration pneumonia

- Metronidazole 500mg po tid
- Amoxicillin 500mg po / iv tid

4. pneumonia in immuno - compromised

- **pneumocystic carnii pneumonia / pcp/**

RX:- co – trimoxazole 480mg 4 tabs tid for 21 days

<u>Viral pneumonia</u>: Treatment is supportive.

Nursing management

- Assess the O_2 therapy effectiveness
- Use relaxation & diversional technique to decrease anxiety
- Provide a quite, restful environment
- Position in semifowler's
- Provide oral care after sputum production
- Humidify inhaled air, steam inhalation
- High fluid intake (3 litter/day)

- Encourage patient to continue on full course of antibiotic therapy
- Encourage patient to obtain adequate rest, good nutrition, and fresh air
- Encourage patient to stop or decrease smoking
- Teach patient to continue coughing and deep breathing

Atelectasis

- It is collapse of an alveolus, a lobule, or larger lung units.

Causes: - Obstruction of a branches by foreign body, or a plug of thick exudates.

-pressure on the lung tissue may be due to pleural effusion, pneumothorax or hemothorax

Pathophysiology

- Reduced alveolar ventilation or any type of blockage.
- Interrupted passage of air to and from the alveoli.
- The trapped alveolar air becomes absorbed into the bloodstream, but outside air cannot replace the absorbed air.
- This portion of lung becomes airless and the alveoli collapse.

C/M: - Marked dyspnea

- Cyanosis
- Pleural pain
- Tachycardia

Mgt:-

- Aspirate the plural effusion & pneumothorax
- Mechanical ventilation may be necessary
- Remove the causes & risk factors

Prevention:

- Encouraging the patient to cough, aspirate secretion, & postural drainage.
- Turn the patient frequently.
- Coughing & deep breathing exercise postoperative pt.

Lung Abscess

- a collection of pus with in lung tissue.

Causes : -Staphylococcus auerus: - is the most common aerobic organism.

  - Anaerobic organisms are much more prevalent

Risk factors:

  - Aspiration
  - Mechanical / functional obstruction of bronchi by tumor, foreign body, or bronchial stenosis
  - Tuberculosis, chest trauma, necrotizing pneumonia, pulmonary embolism

C/M:-

-Fever

- Productive cough of moderate to copious amounts of foul – smelling sputum
- chest pain, dyspnea, weakness, anorexia, weight loss

Dx-History

-Dullness on percussion & decreased or absent breath sounds.

–

- Crepitation
- Chest x – ray, bronchoscope

-Sputum culture

Medical MX: - Admit the patient

- Cloxacillin 500mg IV Qid & change to po after 3 to 4 days
- Postural drainage & chest physiotherapy
- High protein & caloric diet
  - Duration of antibiotic is for 6 weeks
  - Surgical intervention is rare.

Nursing Intervention

- Administer drugs as prescribed & monitor for any adverse effects.
- Chest physiotherapy
- Deep breathing & coughing exercise
- Proper nutrition /high protein & calories diet/ intake encouragement.
- Emotional support.
- Mouth care

Pulmonary Emphysema

- It is an abnormal distention of the air spaces beyond the terminal bronchioles with destruction of the wall of the alveoli & finally lung losses its elasticity.
  - It is the end stage of a process that has progressed slowly for many years.
  - It is irreversible problem

Predisposing Causes: -

- Cigarette smoking : - is the major cause
- Air pollution
- Infection

C/M :

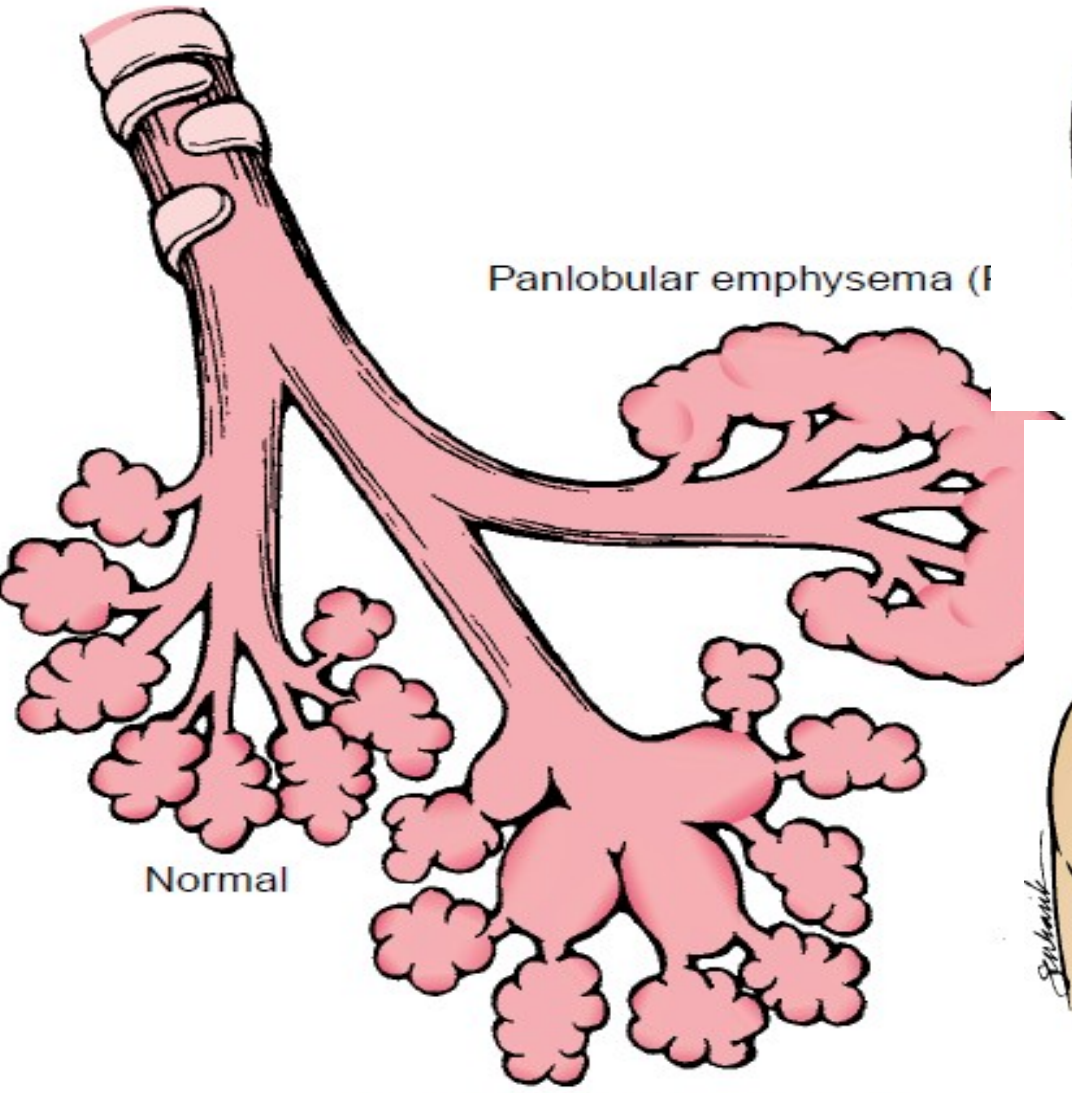
- Dyspnea
- Patient usually has a history of cigarette smoking and history of chronic cough
- Wheezing,
- Teachypnea

- The symptoms are exacerbated with a respiratory infection
- Chronic hyperinflation leads to the “barrel chest
- Hyper resonant on percussion, decreased breath sound
- Anorexia, weight loss, & weakness
- Use of accessory muscles on respiration

DX: - History & physical examination

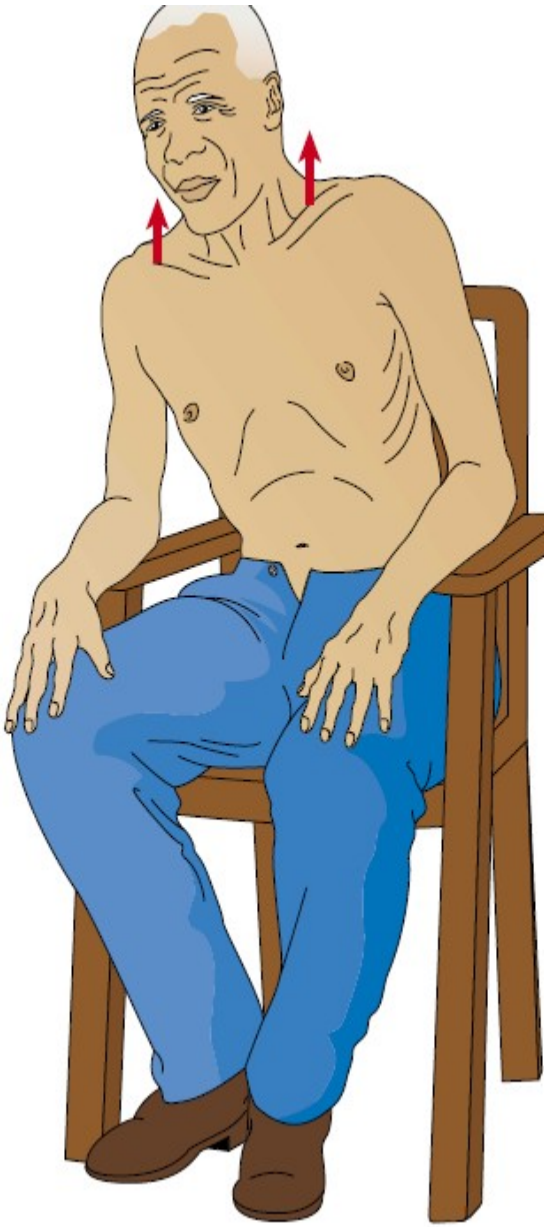
- Chest x – ray

Emphy


Panlobular emphysema (
Normal
Centrilobular emphysema (CLE)







Medical management

1. Bronchodilator:- given to dilate air way & these medications include:
   - Aminophylline, 5mg/kg by slow i.v push over 5 minutes
   - Theophedrin 1 tab po tid
2. Aerosol:
   - Salbutamol aerosol inhalation 2 puff 3 – 4x/day

3. Treatment of infection

- Patients with emphysema are susceptible to lung infections& must be treated at the earliest signs of infection.
- The most common organisms are S.pneumonia & H. influenza.
- Ampicillin 500mg po Qid for 7 – 10 days OR
- Amoxicillin 500mg po Tid for 7 – 10 days OR
- CO – trimoxazole 960mg po bid for 7 days

4. Oxygenation

- Administer oxygen

Asthma

- Asthma is an intermittent, reversible, obstructive air way disease in which the trachea and bronchi respond in a hyper active way to certain stimuli.
- It is manifested by narrowing of the airways, resulting in dyspnea, cough and wheezing.
- The degree of airway narrowing may change either spontaneously or because of therapy.
- Asthma differs from other obstructive lung disease in that it is a reversible processes.

- Acute exacerbations may occur, which last from minutes to hours, interepeted with symptom free periods.

- When asthma and bronchitis occur together, the obstruction is compounded and is called chronic asthmatic bronchitis.

Types of Asthma

1. Extrinsic asthma (allergic asthma)
   - Caused by a known allergen or allergens (eg, dust, pollen, animal dander, food) that releases inflammatory mediators from sensitized mast cells.
   - Most of the allergens are airborne and seasonal
   - Usually seen in persons with family history of allergy and its onset occur during child hood or adolescence.
   - Persons with allergic asthma may also have past medical history of eczema or allergic rhinitis.
   - Skin test for the offending allergens is positive.

2. Intrinsic /idiopathic/ non-allergic asthma

- It is characterized by an absence of clearly defined precipitating factors
- The attacks are frequently precipitated by factors such as infection, weather changes, exercise, exposure to bronchial irritants, and various drugs, including aspirin and other NSAIDS such as phenyl butazone, ibuprofen, and indomethacin.

3. Mixed asthma - is the most common form of asthma.

- It has characteristics of both the allergic & the idiopathic or non-allergic forms.

Clinical Manifestation

- The most common symptoms (triads of symptoms) are:-
  - Cough
  - dyspnea
  - wheezing
- Asthma attacks often occur at night or early in the morning
- Patients prefer sitting position during the attack

- Other symptoms include
  - Chest tightness
  - Prolonged expiration
  - Diaphoresis
- Late signs of poor oxygenation
  - Tachycardia
  - Hypoxemia
  - Cyanosis

DX - complete history & P/E

- Chest x – ray :- over inflated lung
- Sputum (Tenacious, rubbery & whitish) & blood study (IGE)

Complications

- Status asthamticus
- Respiratory failure
- Pneumonia
- Atelectasis

Treatment

I) Acute asthma attacks RX:-

1. Administer concentrated oxygen by mask (6 liters/ min)
2. Rehydrate the patient
3. Drugs Rx –

I / Initial Mx:-

A/ **First line**

- Salbutamol 2 puff & repeat after 20 minute for the first hour

OR

- Aminophylline, 5mg/kg slow iv push over 5 minutes. The same dose could be repeated after 30 minutes.

B/ <u>Alternative</u>

- -adrenaline 0.5ml sc. Repeat after 30 minute to 1 hour if patient doesn't respond.
- - If response to initial therapy is poor, give the following

First line: -

- Aminophylline drip load dose 3-5 mg/kg in dextrose & water over 20 minutes. Then maintenance dose 0.6mg/kg hour in 5% D/W.
- **Plus** Hydrocortisone, i.v 200mg stat
- **AND/ OR** Prednisolone, 40 – 60mg po in divide dose immediately after hydrpcortisone for 5 – 7 days.

II) RX of chronic asthma

1. Intermittent Asthma

- **First Line**: - salbutamol inhalation 2 puff, 3 times a week

  Alternative: - Thephedrine1 tab 3 times per day

2.Persistent Asthma

- **First Line**: - salbutamol inhalation 2 puff 3 times/ day **OR** Theaphedrine 100mg tid
- **Plus** Beclomethasone inhalation puff daily for two weeks.

  OR Prednisalone, 0.5mg, po/ day.

Status Asthmatics

→It is a severe and persistent asthma that does not respond to conventional therapy.

→The attacks can last longer than 24 hours.

→Infection, anxiety, nebulizer abuse, dehydration, increased adrenergic blockage, and non-specific irritants may contribute to these episodes.

→ An acute episode may be precipitated by hypersensitivity to aspirin.

Clinical Manifestations

- The clinical manifestations are the same as those seen in severe asthma:-
- labored breathing
- prolonged exhalation
- engorged neck veins
- Wheezing. However, the extent of wheezing doesn't indicate the severity of the attack. As the obstruction worsens, then wheezing may disappear, and this is frequently sign of impending respiratory failure.

Medical Management

- Initially the patient is treated with short acting beta-adrenergic agonists and corticosteroids
- Oxygen therapy
- IV fluid for hydration

Nursing Intervention

- Check skin turgor for signs of hydration
- Constant monitoring of the patient
- Patient teaching
  - signs and symptoms
  - importance of hydration
  - avoid URTI
  - hazards of over use of drugs

Care of the patient with chronic obstructive pulmonary Disease / COPD/

- **COPD: -** include chronic bronchitis, bronchiectasis, emphysema, & asthma

Nursing Diagnosis

- Impaired gas exchange related to ventilation – perfusion inequality
- Ineffective airway clearance related to broncho constriction, increased mucus production

- Ineffective breathing pattern related to shortness of breath
- Self– care deficits related to fatigue 2^0 to increased work breathing.
- Activity intolerance related due to fatigue, hypoxemia.

Goal(Plan)

- Improvement in gas exchange
- Achievement of airway clearance
- Improvement in breathing pattern
- Independence in self – care activities
- Improvement in activity tolerance
- Compliance with therapeutic program & home care

Nursing Intervention

- Administer bronchodilators as prescribed
- Administer oxygen
- Encourage fluid intake
- Deep breathing exercise
- Perform postural drainage
- Encourage patient to begin to bathe self, dress self & walk
- Support patient in establishing a regular regimen of exercise
- Discuss with patient about drugs

Potential Complication of COPD

1. Atelectasis
2. Pneumothorax
3. Status asthmatics
4. pulmonary hypertention

PLEURISY (PLEURITIS)

- It is an inflammation of both layers of the pleura.
- When the inflamed pleural membranes rub together during respiration(intensified on inspiration), the result is severe, sharp, knife like pain.
- **Pleurisy may develop with**: -
  - Pneumonia
  - URTI, Tuberculosis
  - Pulmonary embolism
  - Chest trauma or simple rib fracture

Two forms of pleurisy:

i. Dry pleurisy(fibrous pleurisy)

ii. Pleurisy with effusion (wet) with increased production of pleural fluid that may result in pleural effusion

Signs & symptoms of dry pleurisy

- Sharp chest pain worst in inspiration, coughing
- Pyrexia, malaise, chills & sneezing
- Dry cough & dyspnea

Signs & symptoms of wet pleurisy(with effusion)

- Pain gradually decreases
- Dyspnea will be more pronounced
- Dry cough
- fatigue

Dx: - C/M

-Chest X- rays

- sputum examination
- pleural fluid analysis
- pleural biopsy

Medical Mx: -

- Treat the underlying disease /pneumonia. . . /
- Analgesics / Indometacin 25mg p.o tid for 1week
- Pleurisy with pleural effusion thoracentesis

Nursing Intervention

- Turn frequently on the affected side to splint the chest wall / this lessen the stretching of the pleura/.
- Emotional support & teaching
- Warm & well ventilated room to reduce cough reflex & chest pain
- Encourage deep breathing & coughing
- Apply heat (hot pad) to the affected site to relive pain

PLEURAL EFFUSION

- is an accumulation of fluid in the pleural space.
- Pleural fluid normally seeps continually through parietal pleural & is reabsorbed by the visceral pleural capillaries & lymphatic system.
- Any condition that interferes with either secretion or drainage leads to pleural effusion.
- The effusion can be composed of a relatively clear fluid, or it can be bloody or purulent.

Causes;

- Increase capillary permeability (e.g. infections or trauma)
- Impair lymphatic function (e.g. lymphatic obstruction due to tumor)
- Increase systemic hydrostatic pressure (e.g. heart failure)

Clinical manifestations

- If the effusion is small (i.e 250 ml.) its presence may be discovered only on chest x-rays
- In large effusion lung expansion may be restricted
- Dyspnea with exertion
- Dry non-productive cough caused by bronchial irritation & mediastinal shift
- Percussion notes may be dull

Diagnosis

- chest x-rays
- Require analysis of the fluid for

- WBC, RBC, Malignant cells

Management

- Identify the cause and manage it
- Thoracentesis is performed to remove fluid.
- Insertion of a small catheter attached to a drainage bottle
- Position in comfortable position
- Care for chest tube

Pulmonary Embolism

- Pulmonary embolism refers to the obstruction of one or more pulmonary arteries by a thrombus that originates somewhere in the venous system or in the right side of the heart.

Risk Factors

❖ **Venous stasis / slowing of blood flow in veins/ which may be due to:-**

- Prolonged immobilization / post operative/
- Prolonged period of sitting
- Varicose veins

- **Hypercoagulability / due to release of tissue thromboplastin often due to:-**
  - -Injury
  - -surgery
  - -Tumor
  - - Increased platelet count
- **Venous Endothelial Disease such as:-**
  - - Thrombophlebitis

❖**Other pre disposing factors include:-**

-Pregnancy

-Oral contraceptive use

C/M: - May be non specific

- Chest pain is the commonest symptom
- Dyspnea is the second most common symptom
- Tachypnea & Haemoptysis
- Tachycardia / rapid & weak pulse
- Cough & diaphoresis
- Syncope & sudden death

** *Multiple small emboli can lodge in the terminal pulmonary arterioles, producing multiple small infarctions of the lungs.*

Diagnostic evaluation

- chest x – ray
- pulmonary angiography
- **Lung scan**
- **Arterial blood gas (ABG) measurements indicate:**

=Arterial hypoxemia (low pa O_2) in massive PE

Prevention:-

- Prevent deep vein thrombosis / active leg exercise to avoid venous stasis, early ambulation /.
- Anticoagulant therapy
- Avoid leaving IV in veins for prolonged periods.
- Elevate legs above level of heart
- Advice against habits that increase venous stasis such as: - Crossing leg

  -Sitting or lying down for prolonged period

  -Wearing constricting clothing

Emergency Interventions

- Oxygen administration to relieve hypoxemia, respiratory distress, & cyanosis
- Open IV infusion
- Catheterize to monitor urinary output
- IV diuretics

Medical Management

1. **Anticogulation therapy**
   - Heparin 5,000 units IV followed by continuous infusion of 1,000 unit per hour is used to prevent recurrence of emboli.
2. **Surgical Intervention**

   -Pulmonary embolectomy

Nursing Intervention

- Prevention:- preventing thrombus formation / ambulation, active & passive exercise to prevent venous stasis /.
- Pain Management
- Oxygen administration
- patient education
- coping with anxiety

Pulmonary Edema

- Abnormal accumulation of fluid in the lung tissue and/or alveolar space.
- Occurs as a result of increased microvascular pressure from abnormal cardiac or kidney function.
- Inadequate left ventricular function causes an increased microvascular pressure, and fluid leak into the interstitial space and the alveoli.
- Hypervolemia or a sudden increase in the intravascular pressure in the lung also cause pulmonary edema.

Clinical Manifestations

- Respiratory distress
- Foam or froth is formed may be coughs up or suctions out blood-tinged secretions, confused or stuporous.
- the pulse oximetry values begin to fall

Management

- Identify and manage cause (usually heart problem)
- Oxygen therapy
- Diuretics may be indicated
- Morphine is administered to reduce anxiety and control pain.
- Intubation and mechanical ventilation if respiratory failure occurs.

Empyema

- It is a collection of purulent liquid / pus/ in the pleural cavity

Etiology

- pneumonitis (bacterial pneumonia)
- Tuberculosis
- Lung abscess
- Carcinoma
- Penetrating chest trauma
- Infection of pleural space
- Thoracic surgery or thoracentesis

Pathophysiology

- At first the pleural fluid is thin, with a low leukocyte count, but it frequently progresses to a fibro purulent stage and, finally, to a stage where it encloses the lung with in a thick exudative membrane (flocculated empyema)

Clinical manifestation

- Fever
- Night sweats
- Pleural pain
- cough
- Dyspnea
- Anorexia
- Weight loss

Diagnosis:

- Chest auscultation demonstrates
  * Decreased or absent breath sounds over the affected area
- Chest percussion reveals
  * Dullness
- Chest x-ray
- Thoracentesis is performed, often under ultra sound guidance

Medical Management

Goal: To drain the pleural cavity & to achieve full expansion of the lung

- Antibiotics for 4 to 6 weeks to sterilize the empyema cavity
- Drainage of the pleural fluid depends on the stage of the disease & is accomplished by one of the following methods:

a) **Thoracentesis (needle** aspiration) with a thin Percutaneous catheter, if the volume is small & the fluid not too purulent or thick

b) **Tube thoracostomy** (chest drainage using a large-diameter intercostals tube attached to water-seal drainage with fibrinolytic agents instilled through the chest tube in patients with flocculated or complicated pleural effusions

c) **Open chest drainage via thoracotomy**, including potential rib resection, to remove the thickened pleura, pus, debris & to remove the underlying diseased pulmonary tissue

Nursing Intervention

- Instruct breathing exercises
- Provide care specific to drainage of pleural fluid
- Give reassurance during thoracentesis
- Administer oxygen and antibiotics as ordered
- Provided chest tube care & use aseptic technique for changing dressings around the tube insertion site

-Watch for respiratory distress or pneumothorax (sudden onset of dyspnea, cyanosis) after thoracentesis

Pulmonary Tuberculosis

- Tuberculosis (TB) is an infectious disease that primarily affects the lung parenchyma but almost all organs can be affected, including the meninges, kidneys, bones, and lymph nodes.
- The infectious agents, ***Mycobacterium tuberculosis*** ,*Mycobacterium bovis and Mycobacterium avium*

TB classification

A. ***Pulmonary TB (PTB): accounts for majority of all TB cases.***
   - Smear-positive PTB: at least 2 smears are positive or one smear positive plus suggestive chest x-ray finding.
   - Smear-negative PTB: If all 3 sputum smears are negative and the patient has suggestive clinical and chest x-ray findings, doesn't respond for broad spectrum antibiotic.
   - Read more……..

B. ***Extra-pulmonary TB (EPTB): TB infecting other than lung***

- Latent TB means a person is infected by TB, but cannot infect others, and is not coughing or sick.
- Active TB means a person infected with TB, is sick and can infect others.

TB Transmission

- Tiny drops sprayed into the air when an infected person coughs, sneezes, or speaks, and another person breathes the air into their lungs containing the TB bacteria.
- ***Factors that facilitate transmission of pulmonary tuberculosis are***
  - Infectivity of the contact (patients with heavy bacterial load)
  - Environment: overcrowding
  - Duration of contact (prolonged exposure)
  - Intimacy (how close the source and the subject are)
  - Host factor (HIV, DM, Immunocompromised, malnutrition, leukimia, renal failure, extreme ages ……..)

Clinical Manifestations

- Low-grade fever,
- Cough: nonproductive at first, mucopurulent sputum or Hemoptysis also may occur later
- Exertional dyspnea.
- Night sweats
- Fatigue
- Weight loss and anorexia.
- Lesion typically involves the apical lobes.
- Pallor, clubbing and inspiratory crepitations

Diagnostic Findings & Management

- Reading assignment(read the current TB national guide line)
- ……
  ……

Lung Cancer / BRONCHOGENIC CARCINOMA/

Risk factors:-

- Tobacco smoker:- lung cancer is 10 times more common in cigarette smokers than nonsmokers.
- Second – Hand smoker: - persons who are involuntarily exposed to tobacco smoke in a closed environment.
- Air pollution
- Occupational Exposure: - chronic exposure to industrial carcinogens, such as arsenic, asbestos, radiation, etc.
- Tuberculosis

C/M: - Cough/ with or without sputum production/

- Wheezing
- Blood – tinged sputum
- Chest pain / late manifestation/
- Dyspnea
- Symptoms of pleural effusion

Diagnostic Evaluation

- Chest x – ray
- Cytologic examination of fresh sputum

-Lung scans

Medical Mx: -

- Surgery: - Surgical resection
- Radiation Therapy
- Chemotherapy

Nursing Intervention

- Remove Secretion from airway
- Deep breathing & coughing exercise
- Oxygen therapy
- Relieve anxiety

CHEST TRAUMA

- Chest trauma is an injury to the chest caused by any form of violence.
- Chest injuries are potentially life-threatening because of:-

(A) Immediate disturbances of cardio respiratory physiology and hemorrhage; and

(B) Later development of infection, damaged lung and thoracic cage.

- Patient with chest trauma may have injuries to multiple organ systems.
- Patient should be examined for intra – abdominal injuries, which must be treated aggressively.

Altered physiology

-In penetrating injuries, some air escapes into the pleural space.

-Negative intrapleural pressure is replaced by atmospheric pressure.

—A loss of negative pressure within the pleural cavity may cause collapse of the lung.

—The change of pressure interferes with expansion of the uninvolved lung, and there is shifting back and forth of the collapsed lung and Mediastinum.

—This shifting interferes with filling of the right side of the heart, lessening cardiac output and causing cardiopulmonary collapse.

Emergency Management

- The order of priority is determined by the clinical status of the patient.
- **Objective: -** To restore normal cardio- respiratory function as quickly as possible.
- This is accomplished by maintaining the **airway, restoring the chest wall integrity, and re-expanding the lung.**

1. **Assess patient's condition physiological state.**

A. Examiner's ear is placed close to patient mouth and nose, allowing him to listen at the airway, watch uncovered chest movements and monitor pulse- this provides a rough estimate of the adequacy of ventilation.

B. Check neck for position of trachea, subcutaneous emphysema and distended neck veins

2. Establish and maintain an open airway

- Aspirate secretions, vomitus and blood from nose and throat.

3. Control hemorrhage

4. Treat for shock. (shock may be due to blood loss, impairment of cardio-respiratory function)

  - Use one or more intravenous infusion lines; obtain blood for baseline studies.
  - Restore blood volume to adequate levels- plasma and electrolyte solutions.
  - Give infusion rapidly.
  - prevent hypovolaemia and circulatory overload.

Types of chest injuries

Pneumothorax

- It is the accumulation of air in the pleural space occurring spontaneously from injury or disease.
- In patients with chest trauma, it is usually the result of a laceration to the lung parenchyma, tracheobronchial tree or esophagus.
- Patient's clinical status depends on the rate of air leakage and size of wound and on previous respiratory condition.

Types of Pneumothorax

a. **Closed pneumothorax** - has no associated external wound. Air from the bronchus, bronchioles, or alveoli escapes in to the pleural space & diminishes lung expansion capacity. The most common form is a spontaneous pneumothorax.

b. **Open pneumothorax** (sucking chest wound) - air enters the pleural space through an opening in the chest wall

c. **Tension pneumothorax**: air can enter on inspiration but air can not escape on expiration

Clinical manifestations:

- Obvious trauma to the chest wall or may not be seen
- Sudden, sharp chest pain & dyspnea
- Shallow rapid respiration
- Cough with or without hemoptysis
- Decreased or absent breath sound
- Tachycardia & decreased BP
- Sweating
- Lack of chest movement on the affected side

Diagnosis

- Sign & symptoms
- CXR - air in the pleural space
  - Mediastinal shift
- P/E - **inspection** - over expansion & rigidity of the affected side
  - **Palpation** - crackling beneath the skin (air in the tissue)
  - **Percussion** - hyperresonance
  - **Auscultation** - decrease or absent breath sounds over the collapsed lung

Management

- Temporary decompression with syringe and needle inserted in to second intercostals space
- Chest tube drainage of pleural space to evacuate air
- Give o_2
- Close the wound

Haemothorax

- Haemothorax is the accumulation of blood in the pleural space.

Cause: - trauma

C/M: - Dyspnea

  - Chest tightness
  - Haemoptysis
  - Signs of hypovolemic shock

Diagnostic tests

- Chest x- ray
- Thoracentesis- aspiration of fluid
- Bronchoscopy

Rx: - Chest tube is inserted to the fourth through sixth intercostals space b/n posterior & anterior line of axilla to drain the blood.

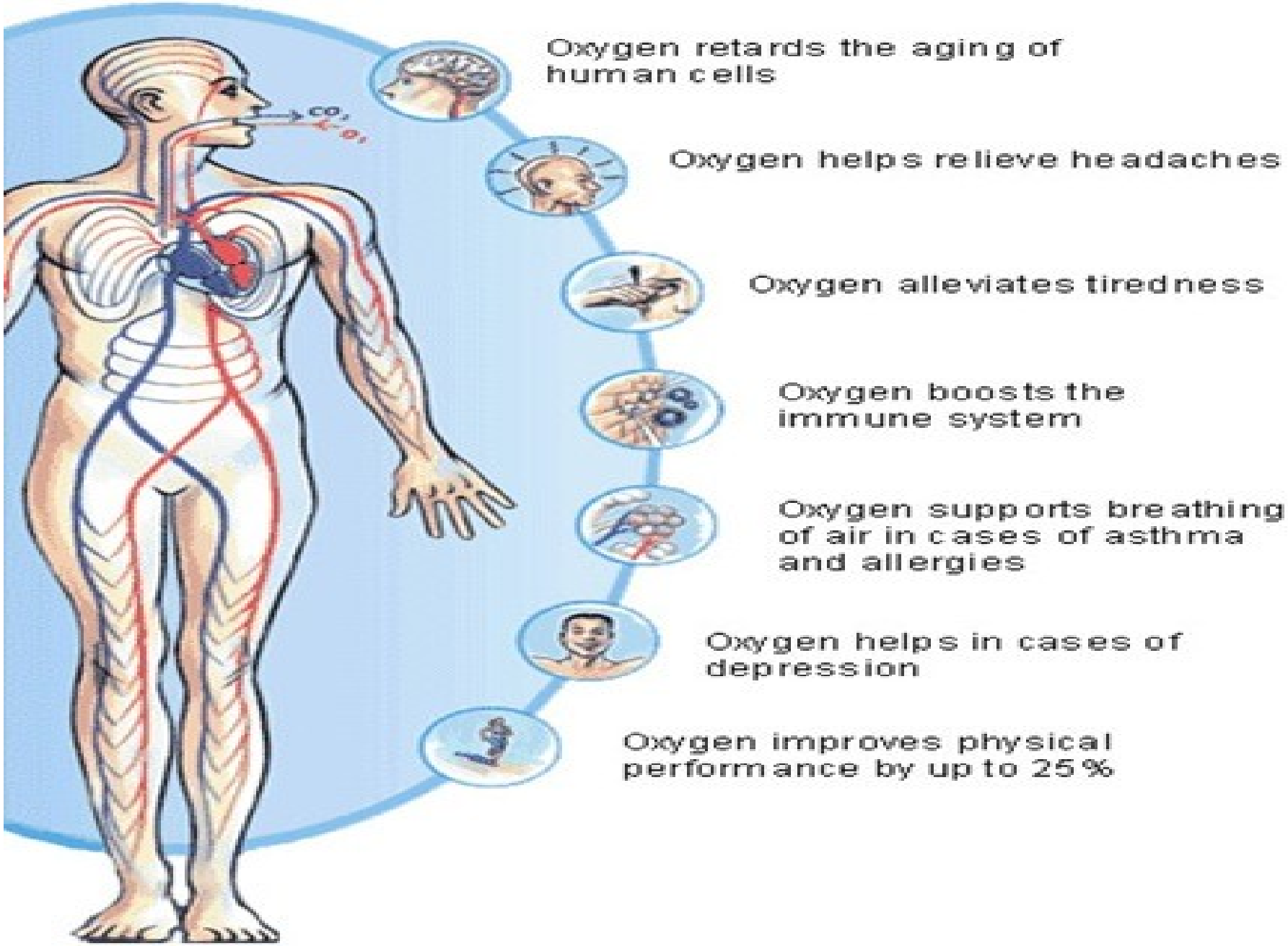
Nursing management of a patient with respiratory disorder

Oxygen administration

Oxygen therapy- is the administration of oxygen as a medical intervention, which can be for a variety of purposes in both chronic and acute patient care.

- Room air only contains 21% oxygen, and increasing the fraction of oxygen in the breathing gas increases the amount of oxygen in the blood.

Benefits of Oxygen Therapy



Additional Benefits of Oxygen Therapy

- Increased clarity
- Relieves nausea
- Can prevent heart failure in people with severe lung disease
- Allows the bodies organs to carry out normal functions

Indications

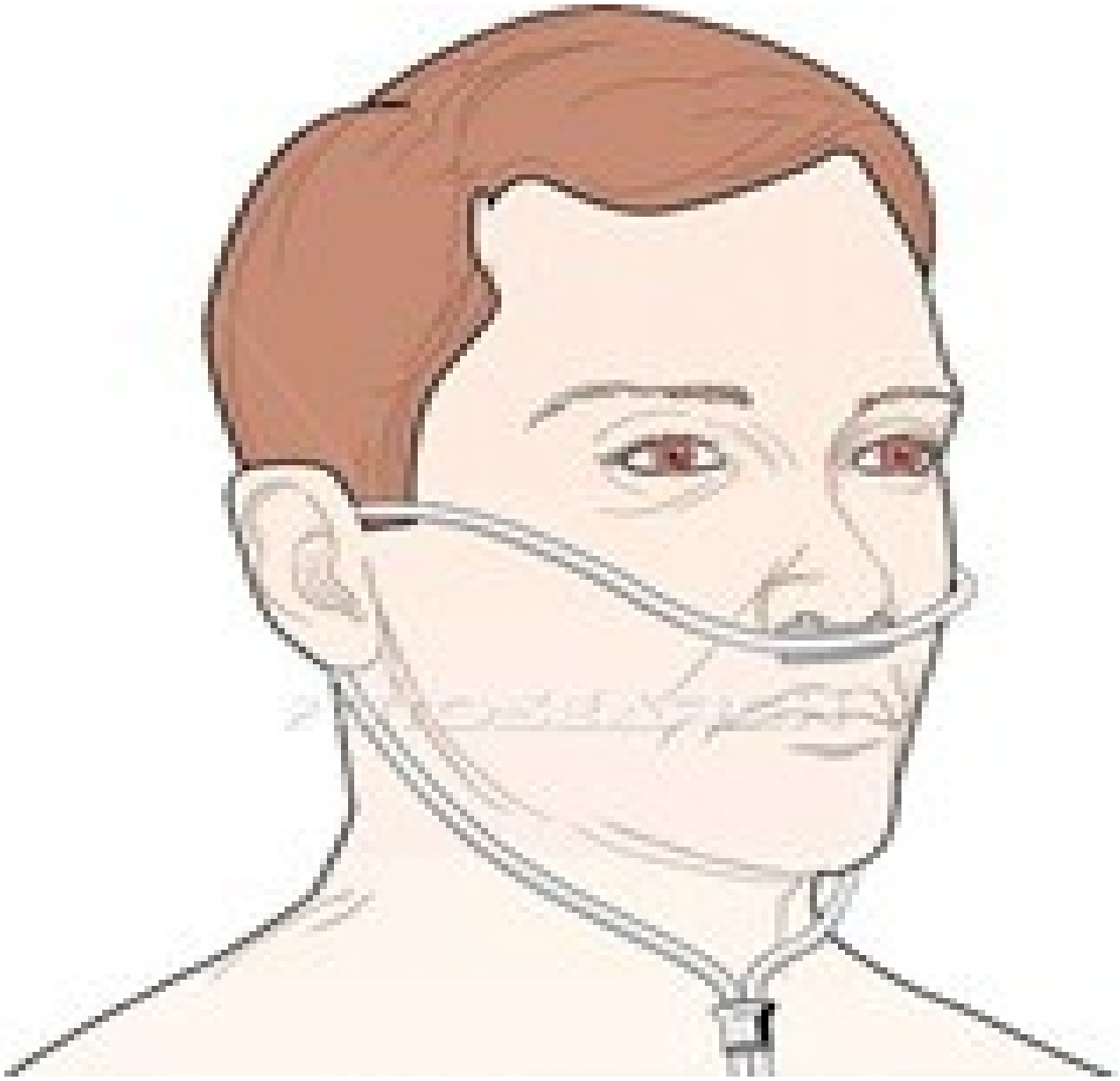
- The main indication for acute oxygen therapy is the presence of tissue hypoxia. This may occur because of:-

1. Arterial hypoxaemia (inadequate arterial oxygen content) or
2. Failure of the oxygen-Haemoglobin transport system Ex. Anemia ,Co poisoning, shock, etc

Any individual with one or more of the following:-

- Cardiac or respiratory arrest
- Hypoxia - diminished blood oxygen levels < 92%
- Acute and chronic hypoxemia (SaO_2 < 92%)

- In emergency condition in resuscitation of major trauma, anaphylaxis, major haemorrhage, active convulsions ,Choking, foreign inhalations ,carbon monoxide poisoning and hypothermia.
- High concentration oxygen is used as home therapy to abort cluster headache attacks, due to its vaso-constrictive effects.
- Oxygen can be administered also in general anesthesia technique





- Oxygen saturation should be monitored by pulse oximetry at least as frequently as observations of vital signs and clearly documented on the patient's observation chart.

Indicators of inadequate oxygenation

- Confusion
- Restlessness progressing to lethargy
- Diaphoresis
- pallor
- tachycardia
- tachypnea

Oxygen toxicity

Signs and symptoms

- Substernal discomfort
- Paresthesia
- Dyspnea, restlessness, fatigue
- Progressive respiratory difficulty, and alveolar infiltrates evident on chest x-ray

Postural Drainage Exercise

- Postural drainage is the use of specific positions so that the force of gravity can assist in the removal of bronchial secretions from the affected bronchioles in to the bronchi and trachea.

Underlying Principles

- The patient is positioned so that the disease area (s) is in a vertical position, and gravity is used to assist drainage of the specific segment(s).
- The position assumed are determined by the location, severity and duration of mucus obstruction.
- The exercises are usually performed two to four times daily, before meals and at bedtime.

Nursing Management

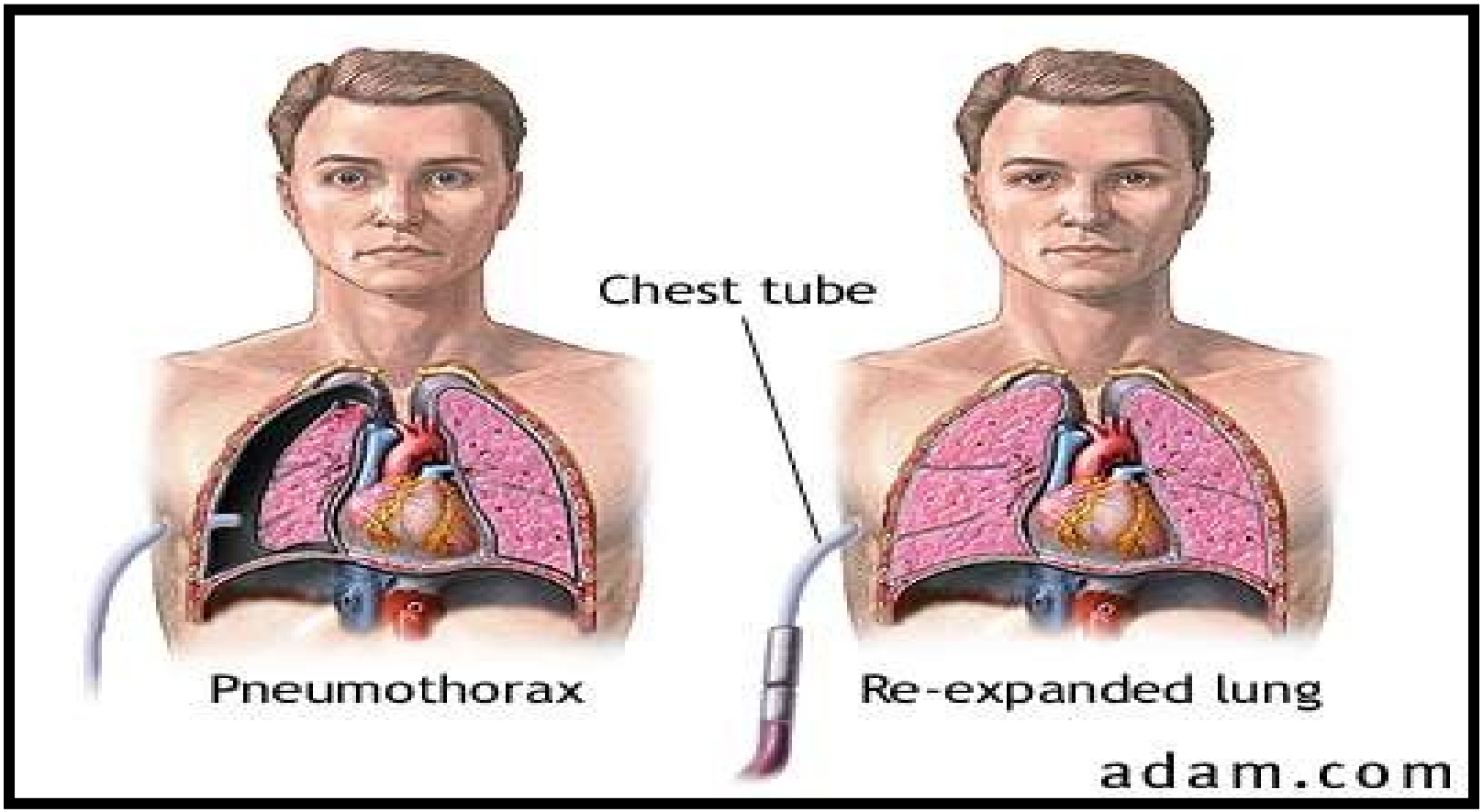
- Make the patient comfortable before the procedure starts and as comfortable as possible while he assumed each position.
- Upper lobes are generally drained by upright positions; lower and middle lobes are drained by head – down position.
- Encourage the patient to cough after he/she has spent the allotted time in each position.

- Evaluate patient’s colour and pulse the first few times he/she performs these exercises.
- Help the patient to brush his teeth and use mouthwash after postural drainage.
- Encourage patient to rest in bed following the procedure.

Reading assignment in your fundamental of nursing course

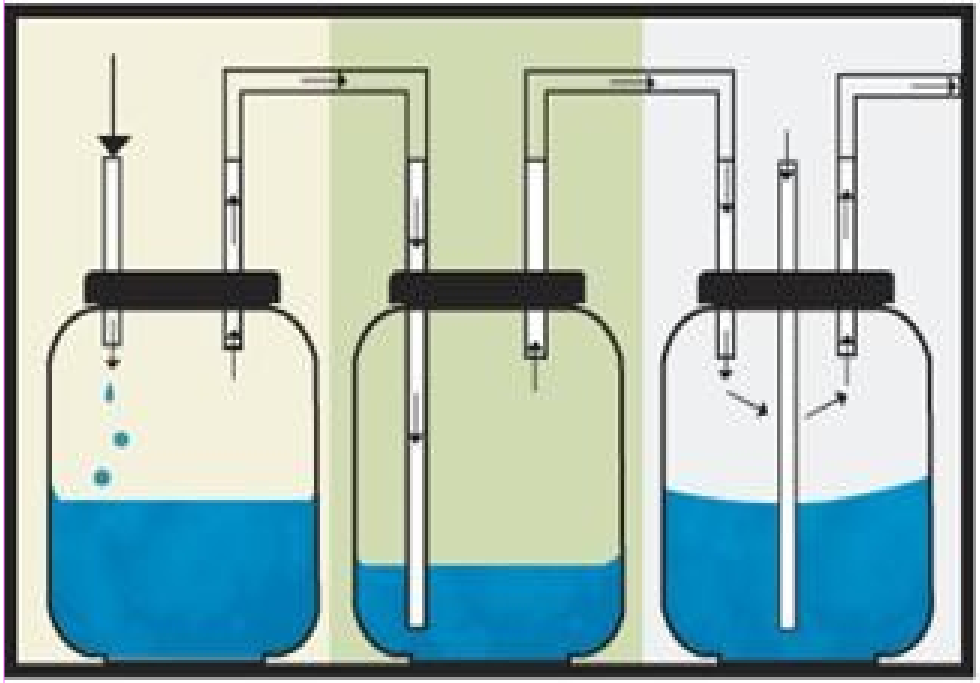
- care of water-seal drainage
- Care of tracheostomy
- Care of pt with thoracentesis

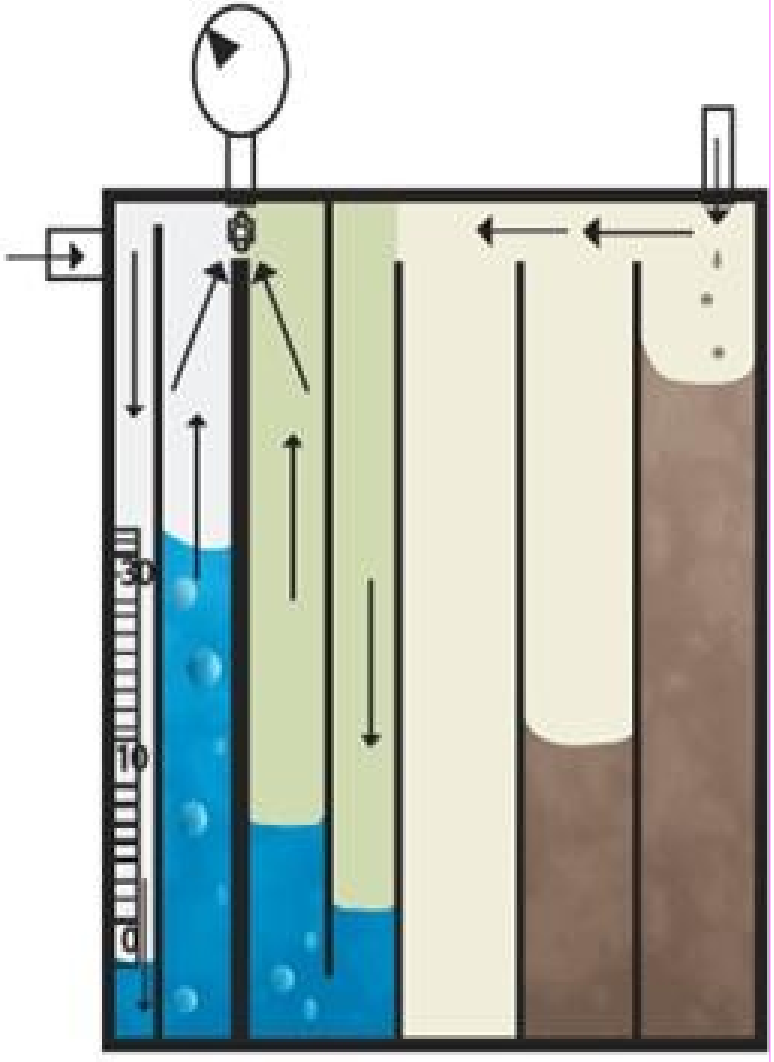
Water-seal chest drainage system



Pleurovac Unit

1. collection chamber
2. water seal
3. suction control chamber

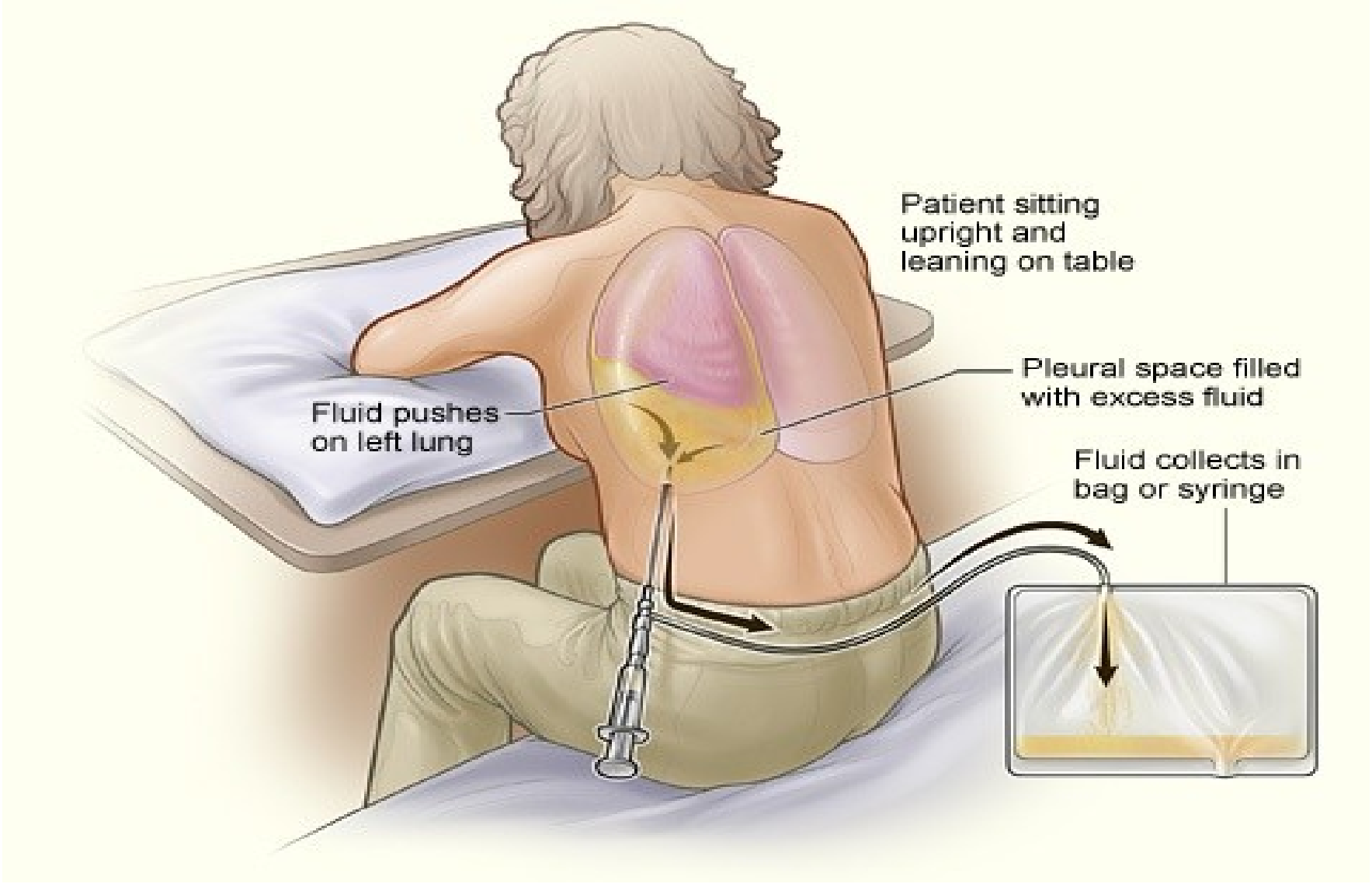




Care of tracheostomy



Care of thoracentesis



Thanks!!!!

End of Respiratory system